

Introduction and Motivation

Dating app profiles are evaluated in seconds. On apps like Hinge, a user's photo stack is the primary filter through which they are judged, often before a single word of their profile is read. As college students, we have heard the frustrations of those around us about how hard it is to find a valid connection on such platforms. Recently, the Senior Scramble, a platform for seniors, where seniors can either choose to "scramble" another person, meaning they match if both people are mutually interested in each other, or "notify", meaning they send the person their interest regardless. There is limited information that you can put on this platform, mainly just a picture, university, and major, emphasizing the importance of the chosen image. Similarly, we have heard friends contemplate for hours about what picture to choose and what captures their life in a way that appeals to others. Even social media apps such as Instagram are often thought of as means of dating as people carefully curate their posts to create appeal.

This project focuses on what kinds of photos people choose to represent themselves and how those choices are perceived by others. We utilized 1,709 posts scraped from r/hingeapp, a subreddit where users voluntarily post their Hinge profiles for public commentary. We built a computer vision pipeline that classifies profile photos along two axes: who is in the frame (solo, selfie, group) and what context the photo depicts (outdoors, gym, everyday life, etc.). The goal is to convert unstructured image content into structured signals that can support downstream profile analysis. Community-sourced review subreddits are a valuable data source for this problem. Unlike lab-collected datasets, these posts are naturalistic: real people uploading the exact photos they are considering, accompanied by unsolicited peer commentary averaging nearly 10 comments per post. The comments capture genuine human judgment about what works and what

does not, largely related to specific image content. Our hypothesis is that there is an association between the classification of images, the person and background context, and the average sentiment of the image.

Background and Related Work

Convolutional neural networks are excellent at image classification tasks, especially when dealing with large amounts of unstructured visual data. While regular CNNs perform well on many tasks, there are often more complex scenes that require the model to capture contextual information beyond simple local features. Krizhevsky, Sutskever, and Hinton's AlexNet demonstrated that deep CNNs trained on large labeled image datasets could outperform prior image classification methods when using methods such as dropout regularization and ReLU activations.¹ Our model incorporates batch normalization after each convolutional block, a technique introduced by Ioffe and Szegedy (2015) that addresses internal covariate shift by normalizing layer inputs at each training step to allow for higher learning rates and acting as a regularizer that can reduce reliance on dropout.² As labels like "everyday" and "outdoors" are far more common than "animal" or "sport/gym," we use PyTorch's weighted random sampler during training to handle class imbalance across our context categories. Research on handling class imbalance in CNN training has found that in almost all cases the best strategy is oversampling the minority class to increase the frequency with which those examples are seen by the model during training.³

¹ [ImageNet Classification with Deep Convolutional Neural Networks](#)

² [\[1502.03167\] Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift](#)

³ [Demystifying PyTorch's WeightedRandomSampler by example | by Chris Hughes | TDS Archive | Medium](#)

The other notable machine learning method utilized in this project is sentiment analysis. Sentiment analysis examines text and outputs a number, typically from 0 to 1, ranking the text on a scale from negative to positive respectively. For our purposes, we use VADER, a rule-based sentiment tool developed by Hutto and Gilbert (2014) and validated specifically on social media text. VADER was shown to outperform several machine learning baselines on Twitter data despite requiring no training. We found it to be particularly well suited to our setting where labeled Reddit comment data is unavailable. VADER also incorporates grammatical and syntactical heuristics that capture phenomena common in online commentary such as the sentiment shift introduced by degree adverbs such as "very" or "kind of" and punctuation patterns like repeated exclamation marks.⁴

Prior research on dating app profile photos has largely centered on facial attractiveness as the dominant predictor of positive reception. Toma and Hancock (2010) found that online daters strategically select photos to manage self-presentation by balancing the desire to appear attractive with the need to appear authentic.⁵ Yet attractiveness alone does not fully account for how profiles are received. Kamelski (2025) analyzed over 73,000 Tinder profile images across 13 countries and found that users construct desirability through distinct visual strategies that extend well beyond physical appearance.⁶ These strategies include lifestyle exhibition and social embeddedness, suggesting that the contextual and compositional features of a photo carry meaningful social information independent of the subject's looks. Our project investigates exactly this possibility and treats photo context and composition as the primary unit of analysis rather than the face itself.

⁴ [VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text](#)

⁵ [Looks and Lies: The Role of Physical Attractiveness in Online Dating Self- Presentation and Deception](#)

⁶ [The Socio-Cultural Anticipation of Attractiveness: Subjectivation through Visual Self-Presentation in Online Dating | Society](#)

Methodology

Data Introduction

Our dataset is sourced from the r/hingeapp subreddit. A community where voluntarily share their Hinge data profiles to receive peer critiques. We scraped 1,709 naturalistic posts containing profile images, post metadata, and community comments. Unlike tightly controlled, lab-collected datasets, these images represent real-world social framing, accompanied by an average of nearly 10 unsolicited comments per post capturing genuine human judgment.

To train our computer vision models, we extracted individual photo crops from the profile images and manually annotated a subset of them. The manual labels were divided along two primary axes: the *person* in the frame (categorized as selfie, solo, or group) and the *context* or background setting (categorized as outdoors, everyday, animal, sport/gym, or other). These labels provided the ground truth necessary for our supervised image classification tasks. The unstructured text data from the comments served as the foundation for our sentiment analysis.

Classification Pipeline

To convert unstructured image content into structured analytical signals, we built a custom Convolutional Neural Network (CNN) framework using Pytorch. We trained two separate CNN models: one to classify the subject of the photo (selfie, solo, group) and another to classify the background context.

To improve model robustness and prevent overfitting, we applied various data augmentation techniques during the training phase. All image crops were resized to 64x64 pixels, and we utilized transformations such as random horizontal flipping, random rotations (up

to 15 degrees), and color jittering to alter brightness and contrast. The CNN architecture utilized sequential convolutional blocks, ReLU activation functions, dropout layers (for regularization), and fully connected linear layers to output the final class probabilities.

The models were optimized using the Adam optimizer and a Cross-Entropy Loss function. Additionally, we implemented a learning rate scheduler (ReduceLROnPlateau) to dynamically reduce the learning rate when validation loss plateaued, ensuring stable convergence. Once trained, the models were used to run inference on the remaining unlabelled images, writing the predicted person and context labels back to our main JSON dataset.

Sentiment Analysis Pipeline

While the computer vision models parsed the visual data, we utilized Natural Language Processing (NLP) to quantify the community’s reaction to each profile. We used VADER (Valence Aware Dictionary and Sentiment Reasoner), a lexicon and rule-based sentiment analysis tool that is specifically attuned to social media text, to evaluate the comments under each post.

VADER processed the texture; critiques and outputted an aggregate, continuous sentiment score for each profile, representing the overall positivity and negativity of the feedback. By merging this continuous sentiment score with the categorical image predictions (e.g, a “solo” person in an “outdoors” context), we created a unified dataset. This integration allowed us to perform downstream statistical modeling—such as Logistic Regression—to evaluate which combinations of photo subjects and contexts were most strongly associated with positive peer reception.

Results

The results of the image classification models reveal significant challenges in both person and context categorization, primarily due to class imbalances and insufficient input resolution. In the person classification task, the solo category achieved the highest precision (0.75), likely because it dominates the training set with 941 samples, allowing the model to learn its features more effectively than the rare selfie (98) or group (154) classes. On the other hand, the selfie category performed poorly in precision (0.09), while the sport_gym context category was the least accurate, achieving a 0.00 F1-score. This failure suggests the model struggled to differentiate gym environments from other indoor settings, perhaps because there weren't many visible and distinct features at a 64 x 64 resolution.

There are many things to consider when evaluating the poor performance of this classification model, with validation accuracy hovering around 50% for person and 32% for context. First, the image size of 64 is probably too small for the model to see small but notable details, such as the difference between a person holding a phone for a selfie versus standing alone in a solo picture. Second, the class imbalance is noteworthy. The model likely developed a bias toward the "solo" and "outdoors" classes because they appeared most frequently. This can be seen by the "other" context category having a high recall 0.72 but low precision 0.15, indicating the model likely labeled many images it couldn't confidently distinguish as 'Other'.

Broadly, across all categories, sentiment scores are positive and typically between 0.45 and 0.58. Being that the sentiment analysis model that was used, VADER sentiment analyzer, has a scoring scale of -1 to 1, these scores indicate that the images used from the Hinge profile posts tend to receive strongly positive feedback. The general category with the lowest score was the

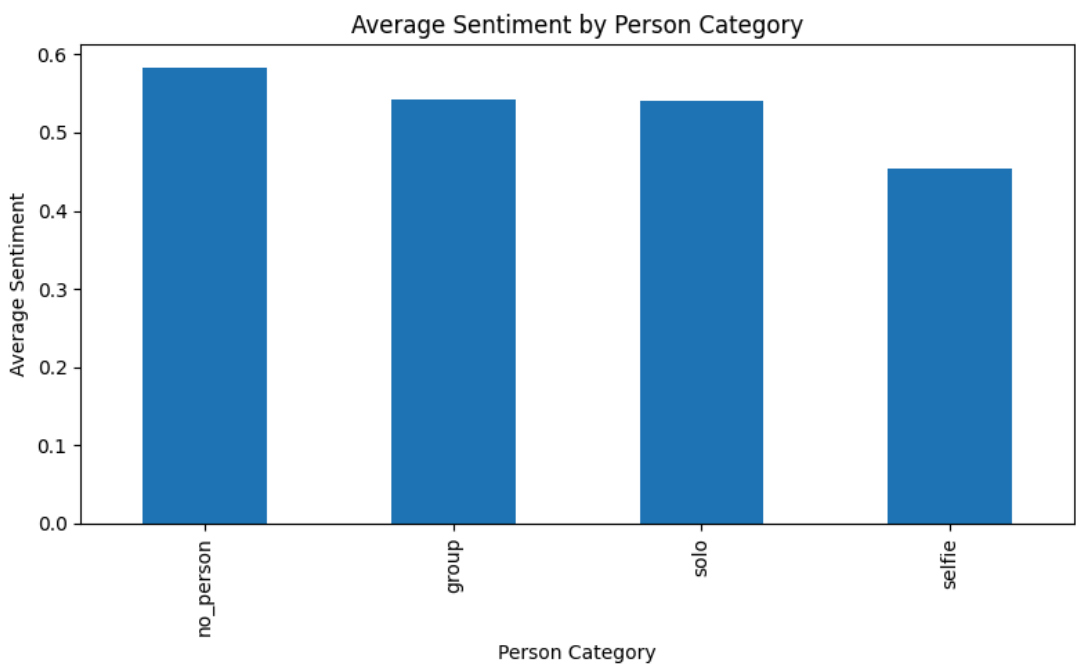
selfies category with a score of 0.45 but contextually, on the scale this is still a very favorable score. Then, the question becomes, while all these categories may be viewed positively, which ones perform better than others? To help contextualize our findings, see the tables with image counts of each person and context category below.

Image Count By Person Category	
Solo	941
Group	154
Selfie	98
No Person	26

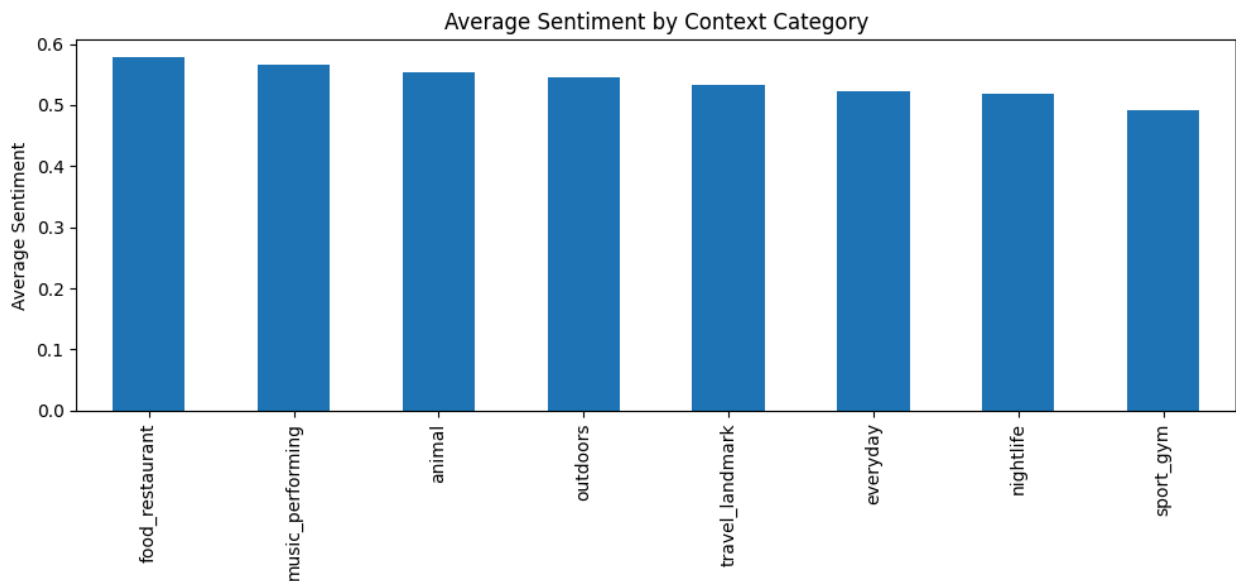
Image Count by Context Category	
Food/Restaurant	44
Music/Performing	34
Animal	89
Outdoors	512
Travel/Landmark	16
Everyday	409
Nightlife	30
Sport/Gym	85

Among the people categories, the most notable pattern is that images without people achieve the highest average sentiment (0.583), followed by group (0.542) and solo (0.541)

images, while selfies perform the worst (0.454). Socially, selfies being the lowest ranking ‘people’ category aligns with current social media preferences. Selfies can be perceived as self-absorbed and uninteresting compared to other more creative or socially involved images. The highest ranking category (no person) corroborates this idea. Images without people can include landscapes and interesting objects that invite curiosity and receive broader appreciation than a personal selfie. The graph below shows the average sentiment across all person categories.

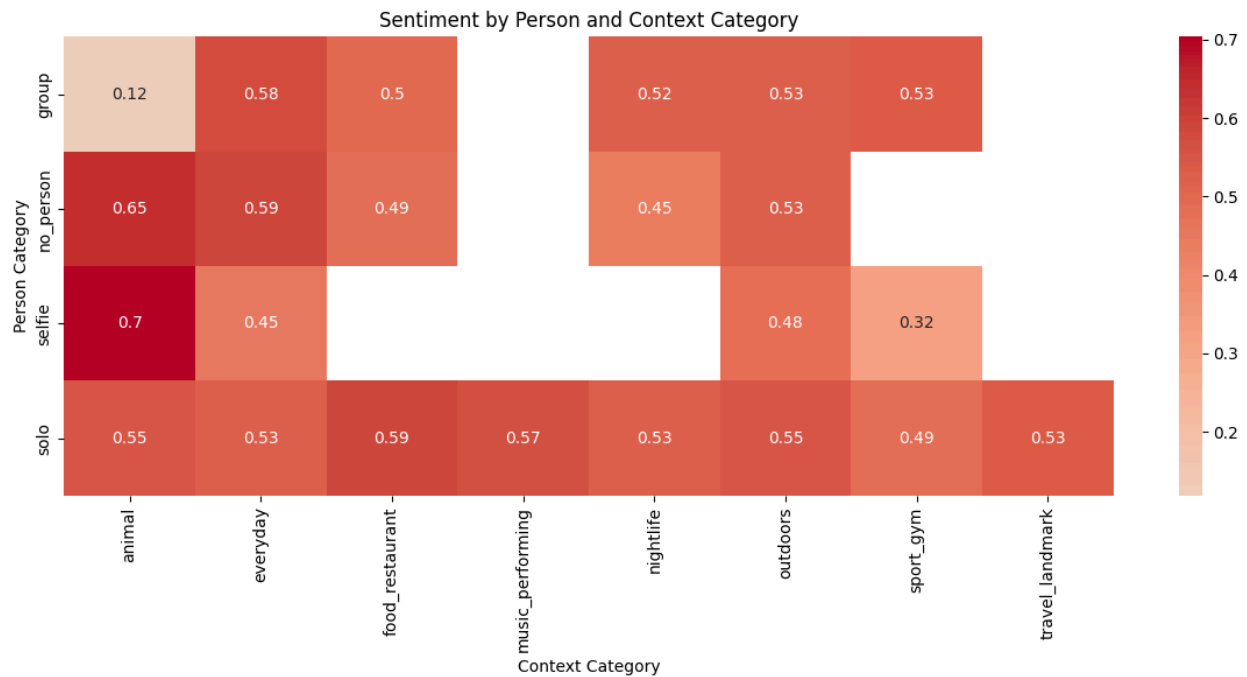


In the context categories we see a continuation of overall positive engagement. Interestingly, the categories with the highest sentiments were food/restaurant (0.578),



music/performing (0.566), and animal (0.555). One possible explanation is that these categories convey to other users that the person invests in engaging experiences such as trying new cuisine or attending live events. These are typically celebratory events that can evoke happiness. Still, the other categories don't fall far behind in their scores. Nightlife (0.518) is second to last with Sport/gym (0.491) under it. These lower sentiments could be explained by the fact that these activities can receive mixed reactions depending on whether other users approve of the lifestyle, as in the case of nightlife. Others may also question whether the person included a sport/gym image to share their hobby or to flaunt their efforts. The graph below shows the average sentiment across all context categories.

The interaction results provide the most nuanced insights. Some combinations perform particularly well such as solo + food/restaurant (0.588, n=39), group + everyday (0.577, n=62), and solo + music/performing (0.566, n=34). These combined categories have a score higher than their respective person category scores. This suggests that context can amplify the reception of an image regardless of who is in it. Solo images, third highest sentiment for people categories, performed strongly in combination with context categories related to experiences (food, music). However, the lowest general sentiment scores also arose from combinations. For example, selfie + sport/gym (0.316, n=2) and group + animal (0.119, n=2). These low scores should not be surprising given that they have very small sample sizes. With only 1–2 observations, these averages are not reliable or generalizable without significantly expanding the sample size. This highlights an important limitation to our usage of this unsystematic dataset. Combinations with low observations can produce misleading results. Lastly, as you can see in the heatmap detailing sentiment scores in combinations of person and context categories, some categories did not have observed combinations and those are left with an empty cell.



Implications, Limitations, and what we would do if we had more time

Initially, we hoped that the prominence of dating apps in 2026 would lead to an influx of datasets about users and their behaviors on these apps. After all, our initial research question was investigating the relationship between the types of photos users included in their dating app profiles and the number of matches they received. However, it seems that these apps have taken the responsibility of caring for their users' privacy seriously. We were unable to find publicly available data containing user profiles and their engagement patterns.

One of the initial datasets we found on Kaggle was titled ‘Speed Dating Experiment’⁷ (Montoya 2006). The description mentioned that the data was gathered from various in-person speed dating events sponsored by the Columbia Business School in the early two-thousands. Participants were asked a series of questions ranking their personal attributes and their romantic

⁷ [Speed Dating Experiment](#)

preferences in a partner. But, while the size of the dataset seemed promising (8000+ observations), the dataset did not appear to be prepared for use by third party members. Among the 120+ features in the dataset, many had uninterpretable and ambiguous headers such as 'expected_happy_with_sd_people', 'd_age', and 'exercise.' Without a detailed description explaining the context behind these variables, we were left to guess their meaning. Because of the ambiguity of the dataset, we decided to seek help.

After meeting with our teaching assistant, as a group, we decided to pivot our project from analyzing dating app profiles to evaluating social media profiles such as those on Instagram. The goal was still to classify users' images into categories and gauge others' engagement with the image through likes counts and comments. Quickly we realized this method demanded a lot more caution and attention to detail from us. Which profiles would we select to scrape? How many? Should we only include influencers or users with high expected engagement to ensure that we have enough data measuring engagement? What impact does only including images from out-of-touch celebrities with unrealistic lifestyles have on the generalization of our results? If we include normal, average civilians in our dataset, is it ethical to include potentially identifying information about unconsenting users into a machine learning pipeline? The burden of these questions ultimately led us back to square one.

Then came our saving grace. The subreddit r/hingeapp describes itself as “a community for discussing the Hinge dating app. Request a profile review, ask for advice, get help, or share your experiences with Hinge.”⁸ Reddit users voluntarily post images including sections of their Hinge profile and openly request feedback from the community about improvements they can make to their profiles to increase their matches. The subreddit included all the necessary

⁸ <https://www.reddit.com/r/hingeapp/>

components we required in a dataset: images from users seeking romantic interactions, and articulable comments from other users assessing the merit of that style of image in improving these interactions. Now, our research question was adapted to analyze the relationship between types of photos in dating app profiles and the efficiency of these styles of photos in increasing the possibility of a match.

Still, there was much to do to prepare this data for a machine learning project. Using a Convolutional Neural Network classification model to label images in their people (solo, group, selfie) and context categories (outdoors, everyday, animal, gym, other) resulted in blatant and extensive errors. Using these labels in the training dataset would be improper scientific practice and lead to an impractical model. So, we meticulously labeled the 1000+ images used in the training dataset by hand. Even still, the accuracy and precision metrics of our classification model are noticeably poor. To improve the model in the future, several adjustments should be implemented to handle the complexity of the dataset. Increasing the image size to at least 224 x 224 would provide the CNN with the necessary spatial resolution to identify key contextual objects. Additionally, to combat the imbalance, class weighting could be added to the Cross Entropy Loss function. We could also make use of transfer learning by using a pre-trained model like ResNet to improve the accuracy in classification even with our smaller dataset.

On the other hand, this project is more complex than just image classification. Under each Hinge profile post, there were comments with other users' feedback on the images. These comments were inputted to a sentiment analysis model to measure whether this category of image receives positive or negative feedback. Each post only has one comment section but includes multiple images of the user's profile, and these images each have a different people and context category. This means that negative comments about a selfie included in the user's array

of images can decrease the sentiment attributed to the group picture also featured in the post. The model did not discriminate the comments based on which image the comment was referring to. This is another element of our project that with extended time and resources we would improve by matching subsets of text in the comment with the corresponding image it was referring to before running the model.

Conclusion

In this project, we successfully developed an end-to-end machine learning pipeline capable of parsing, classifying, and evaluating the social context of dating app pictures. Prior research has historically focused on evaluating photos based purely on facial attractiveness, which neglects the crucial contextual signals—such as social status, hobbies, and setting—that users convey to potential matches. By leveraging real-world, naturalistic data from r/hingeapp, we addressed this gap.

Our dual CNN approach effectively transformed disorganized profile images into structured analyzable features representing both the subject and their environment. By correlating these visual classifications with VADER-generated sentiment scores derived from peer critiques, we established a quantifiable link between profile composition and human perception. Ultimately, this pipeline proves that community-sourced critique forums can be transformed into robust datasets, offering scalable and data-driven guidance for users looking to optimize their digital first impressions.